BRNO UNIVERSITY OF TECHNOLOGY FACULTY OF ELECTRICAL ENGINEERING AND COMMUNICATION

Robotics and AI

MPC-MAP Assignment 1

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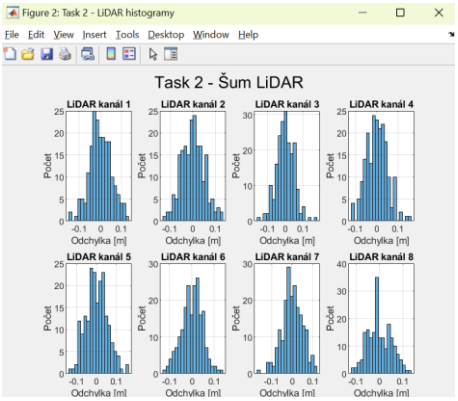
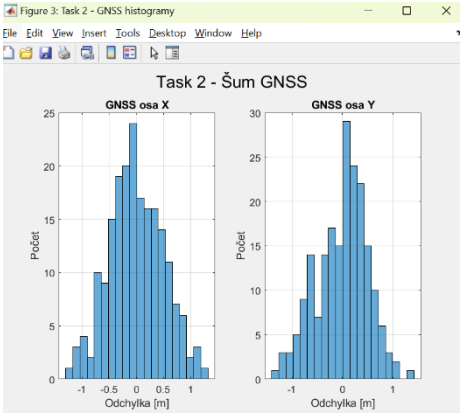
Task 2

Robot byl na fixní pozici, bylo sbíráno 200 vzorků. LiDAR: σ ≈ 0,05 m, GNSS: σ ≈ 0,5 m. Šum LiDARu a GNSS je konzistentní napříč všemi kanály a osami. Pro LiDAR byla použita mapa indoor_1, pro GNSS byla použita mapa mixed_1.





Task 3

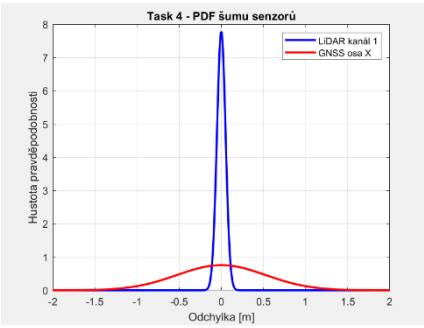
Pomocí funkce cov() v MATLABu byly sestaveny matice z nasbíraných vzorků. LiDAR: matice 8×8, GNSS: matice 2×2. Hodnoty na hlavní diagonále odpovídají rozptylu σ².

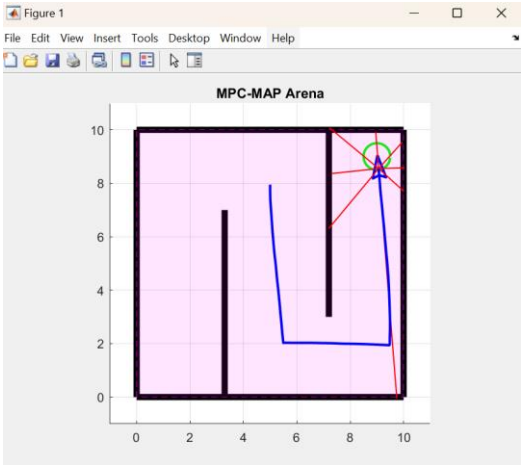
```
TASK 3
LiDAR
    0.0026   -0.0001         0         0         0    0.0000    0.0002   -0.0003
   -0.0001    0.0033         0         0         0    0.0003   -0.0001   -0.0001
         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0
    0.0000    0.0003         0         0         0    0.0027    0.0003   -0.0003
    0.0002   -0.0001         0         0         0    0.0003    0.0024   -0.0001
   -0.0003   -0.0001         0         0         0   -0.0003   -0.0001    0.0022

    0.0026    0.0033         0         0         0    0.0027    0.0024    0.0022

Ocekavane hodnoty (sigma^2):
    0.0026    0.0033       NaN       NaN       NaN    0.0027    0.0024    0.0022
```

```
GNSS
    0.2780   -0.0227
   -0.0227    0.2518

Ocekavane hodnoty (sigma^2)
    0.2780    0.2518
```

Task 4

LiDAR (modrá) pro kanál 1 má výrazně užší PDF než GNSS osa x (červená).



Task 5

```
function [public_vars] = plan_motion(read_only_vars, public_vars)
c = read_only_vars.counter;
if c <= 120
    public_vars.motion_vector = [0.5, 0.5];
elseif c <= 136
    public_vars.motion_vector = [0.08, -0.1];
elseif c <= 216
    public_vars.motion_vector = [0.5, 0.5];
elseif c <= 232
    public_vars.motion_vector = [0.1, -0.1];
elseif c <= 375
    public_vars.motion_vector = [0.5, 0.5];
else
    public_vars.motion_vector = [0.0, 0.0];
end
target = get_target(public_vars.estimated_pose, public_vars.path);
end
```